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An analytical framework based on the recency, frequency, and monetary model and time series clustering techniques for dynamic segmentation

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ABSTRACT

Nowadays, banks use data mining and business intelligence tools and techniques to analyze their customers' behavior. Customer segmentation is a widely adopted analytical tool to identify distinct groups of customers. Most studies have adopted a static segmentation approach, leading to missing important trends and patterns. In this study, we propose a framework that represents each customer behavior as a time-series sequence of the Recency, Frequency, and Monetary variables, and then exploits time-series clustering algorithms to carry out customer segmentation. The framework consists of state-of-the-art clustering algorithms, including hierarchical, spectral, and *k*-shape algorithms to divide customers into homogeneous groups and implement point of sale devices transaction data of grocery and appliance retailers. We divide customers into four segments, analyze the behavioral trends, and provide marketing suggestions for each segment. Our results show that, in terms of the computed validity indices, the best clustering model for grocery retailers is reached using hierarchical clustering with the Complexity-Invariant Distance measure, and for appliance retailers, the best segmentation is achieved by applying the spectral clustering with the Complexity-Invariant Distance measure.

1. Introduction

Customer relationship management (CRM) has become increasingly important due to the intense competition among firms in different industries (Abbasimehr & Shabani, 2019; Guerola-Navarro et al., 2020; Kumar & Reinartz, 2018; Zerbino et al., 2018). According to Ngai et al. (2009), CRM activities are divided into two distinct categories; operational and analytical CRM. Analytical CRM aims to use data mining (Han et al., 2011) and business intelligence techniques (Ain et al., 2019; Božič & Dimovski, 2019; Rasha & Walid, 2020) to gain insights into customers (Ngai et al., 2009). Customer segmentation is a popular tool for identifying different customer groups with similar needs and behaviors (Manjunath & Kashef, 2021; Parvaneh et al., 2014). The Recency, Frequency, and Monetary (RFM) model is a widely used approach to conduct behavioral customer segmentation (Abbasimehr & Shabani, 2021; Hajipour & Esfahani, 2019; Hughes, 2011; Parvaneh et al., 2014). Although data utilized for customer segmentation based on the RFM model is a time-stamped transaction data generated by sale systems, most previous studies have employed a static segmentation approach by aggregating customer transaction data over a fixed period (see for example Djurisic et al. (2020), Dogan et al. (2018), Mahdiraji et al. (2021), Parvaneh et al. (2014), Peker et al. (2017), Wei et al. (2019)). The main drawback of the static segmentation approaches is that they cannot model customers' dynamic behavior and discover meaningful trends and patterns (Akhondzadeh-Noughabi & Albadvi, 2015; Mosaddegh et al., 2021; Seret et al., 2014). Therefore, to extract useful patterns about customer behavior, dynamic segmentation approaches should not be overlooked (Akhondzadeh-Noughabi & Albadvi, 2015; Böttcher et al., 2009; Hosseini & Shabani, 2015; Montoya et al., 2010; Seret et al., 2014). Although some previous studies have attempted to capture customer's dynamics using clustering, they have difficulties predicting customers' future behavior as their proposed approaches are mainly descriptive and allow an exploratory understanding of customer behavior.

This study proposes a framework that incorporates the time dimension of customer behavior by representing time series to capture customer dynamics. Time-series sequences are the best way to track customer behavior over time as they preserve temporal order of observations (Abbasimehr & Shabani, 2019; Yanovitzky & VanLear, 2008). In the proposed framework, RFM attributes are extracted from customers' transaction data. Then, customer segmentation is carried out using time series clustering. The resulting clusters' quality in a time series clustering task depends on choosing a suitable distance measure and an effective clustering algorithm (Paparrizos & Gravano, 2017). In this framework, state-of-the-art distance measures such as

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dynamic time warping (DTW), complexity-invariant dissimilarity (CID), temporal correlation coefficient (CORT) (Montero & Vilar, 2014) and Shape-based distance (Paparrizos & Gravano, 2017) and some powerful time-series clustering algorithms including hierarchical clustering (Tan et al., 2018), spectral clustering (Ng et al., 2001), and *k*-shape clustering (Paparrizos & Gravano, 2017) are combined to obtain the best quality clustering results. After obtaining the best clustering results, the centroid time series for each cluster is extracted to represent it. Afterward, each segment is labeled and its behavioral trends are analyzed to reveal its dominant patterns. To demonstrate the usefulness of the framework, we conduct a case study on a bank's merchants who use point of sale (POS) devices to process consumer card payments. Our case study is in the context of business-to-business (B2B) as it considers merchants as business customers of the bank (Houjeir & Brennan, 2017). The results of the case study reveal useful patterns about the behaviors of the bank's customers.

The rest of this paper is organized as follows: In Section 2, we provide a thorough description of the related work. Section 3 introduces the different techniques used throughout the paper and then describes the proposed framework for dynamic customer behavior analysis. In Section 4, the results of the proposed framework on a case study are reported, and the discovered behavior patterns are analyzed. The managerial implications are provided in Section 5, and Section 6 concludes the paper and suggests some future studies.

2. Related work

Customer behavior analysis often has been performed through customer segmentation. Past studies in customer segmentation are mainly categorized into static and dynamic segmentation (Akhondzadeh-Noughabi & Albadvi, 2015). The main disadvantage of static approaches is that they are valid at a specific time point and miss significant customer behavior trends (Akhondzadeh-Noughabi & Albadvi, 2015; Mosaddegh et al., 2021; Seret et al., 2014). To tackle this problem, some approaches have been proposed that consider the dynamics of customer behavior in the analysis.

Table 1 provides the distinguishing characteristics of some of the past studies that accomplished dynamic behavior analysis, including their contributions, techniques used, modeling variables, and contexts.

In the current literature, there are mainly two approaches for dealing with customer dynamic (Akhondzadeh-Noughabi & Albadvi, 2015).

In the first category, studies aim to mine segment changes over time (e.g., Böttcher et al. (2009), Mark et al. (2013)). These studies collect customers data at time point $\{t_1, \ldots, t_n\}$ and perform clustering at each time point $\{c_1, c_2, \ldots, c_n\}$ and label them. Next, they identify changes in segments or cluster transition over time concerning the content or structural changes (Blocker & Flint, 2007). Most current studies in this category have investigated only the content change via frequent itemset and association rule mining.

In the second category, studies concentrate on tracking customer shifts between segments over time (see Akhondzadeh-Noughabi and Albadvi (2015), Lemmens et al. (2012), Mark et al. (2013), Mosaddegh et al. (2019, 2021), Netzer et al. (2008)). Two primary approaches have been utilized in this category of studies. (1) Employing Hidden Markov models (HMMs) for modeling customer transition between segments: these studies capture the evolution in customer–firm relationships and investigate the customers' behavioral changes over time (Lemmens et al., 2012; Mark et al., 2013; Netzer et al., 2008). Mark et al. (2013) classify customer shifts between segments into two types: customer conversion, that is the evolution of customer from one type of relationship to another type, and customer switching (attrition), that is the loss of customer by a firm. (2) Employing association rule mining to extract the dominant patterns in customer shift (Akhondzadeh-Noughabi & Albadvi, 2015; Mosaddegh et al., 2019, 2021): these studies have utilized the association rule mining to capture the dominant transition patterns that customers follows.

The current literature that has considered the dynamics of customers is primarily based on descriptive methods such as association rule mining, which extracts some patterns regarding changes in customer segment and customer behavior over time. Unfortunately, the literature has an obvious drawback as the majority of the studies do not incorporate the time dimension of individual customer behavior. They mainly have monitored customer behavior at the segment level by monitoring content and structural changes in customer segments over time or extracting customer shift patterns.

However, in this study, we propose to capture the dynamic behavior of a customer by representing it as a time series. Time series data are different from non-time series one for the reason that observations have a temporal order and the analysis of such data capture the dynamic behavior of a customer over time effectively. In addition, it also allows predicting the future behavior of the customer, which is considered as another key advantage of modeling customer behavior as time series (Yanovitzky & VanLear, 2008).

Based on Table 1 clustering and association rule mining are the widely used data mining techniques in the previous studies for dynamic segmentation, which are exploratory methods. Subsequently, the resulting models of such approaches are descriptive in nature, which makes their performance measurement difficult. The *k*-means algorithm is another technique that has been extensively used for customer behavior analysis. However, as emphasized in Montero and Vilar (2014), *k*-means is more suitable for non-time series data; nonetheless, its performance may not be acceptable in terms of cluster validity index for clustering time-series data. This is because the *k*-means algorithm assigns a data point to a cluster considering its distance to the centroid point of that cluster. Therefore, the performance of clustering mainly depends on defining each cluster's centroid point, which in this case, each data point is a time series data. Computing the centroid points of *n* time series is a challenging issue as it requires aligning multiple temporal data simultaneously. However, *k*-means has been developed initially with the Euclidean distance, which does not consider this issue.

3. Methodology

This section firstly introduces the time series clustering techniques as well as time series distance metrics. Then, it integrates these concepts to formulate the proposed methodology.

3.1. Time series clustering

Clustering is a challenging task, and the quality of clustering results depends on two critical decisions; the choice of the clustering algorithm and the selection of a suitable distance measure (Paparrizos & Gravano, 2016). Therefore, it is essential to consider several clustering algorithms and distance measures to obtain well departed and homogeneous clusters. In this subsection, we describe the clustering algorithms and the distance measures that are incorporated in the proposed framework.

According to Batista et al. (2014) and Paparrizos and Gravano (2015), the three most popular clustering methods are hierarchical, spectral, and partitional clustering. Therefore, in this study, we have performed clustering using the three representatives of these clustering types combined with the most competitive distance measures to find the best clustering model. For hierarchical clustering, we used the agglomerative hierarchical clustering (Montero & Vilar, 2014), for spectral clustering, we adapted the popular normalized spectral clustering (Ng et al., 2001), and finally, for partitional type, we used the novel *k*-shape algorithm (Ng et al., 2001; Yang et al., 2017). The main advantage of the hierarchical clustering method is that it does not need a predefined number of clusters. Also, it does not need to compute cluster centroid, which is a challenging task in the time series context. Also, it provides the ability to visualize the results of clustering through dendrogram (Montero & Vilar, 2014; Wang et al., 2019). Also,

Table 1
Summary of the past studies that conducted dynamic behavior analysis

Study	Contribution	Techniques used	Modeling variables	Context
(Böttcher et al., 2009)	Proposed a system that tracks how customer behavior changes over time	Frequent itemset mining	Survey data, Network usage data	Telecommunication
(Montoya et al., 2010)	Proposed a Hidden Markov Model (HMM)-based method to capture the dynamic behavior of physicians in the prescription of drugs	HMM	Prescriptions data, detailing and sampling data	Pharmaceutical industry
(Mark et al., 2013)	Developed a model to investigate how relationships between customers and retailers change over time	HMM	Sale transaction data	Apparel and household goods retailing
(Seret et al., 2014)	Discovered frequent customer behavior trajectories through time to understand customer's dynamics	Clustering (Prioritized-self-organizing map (P-SOM), *k*-means), Sequence Mining (Generalized sequential pattern)	Demographic data, customer ordering transactions	Ticketing
(Akhondzadeh-Noughabi & Albadvi, 2015)	Identified customer shift patterns by utilizing a combined approach of customer segmentation and sequential rule mining	Clustering (*k*-means), Sequence mining (Generalized sequential pattern)	RFM	Telecommunication
(Hosseini & Shabani, 2015)	Tracked customer behavior over time by segmentation of customers in each time point then extracting the sequence of customers' label	Clustering (*k*-means)	RFM	Banking
(Zhang et al., 2016)	Combined multi-state model (MSM) and HMM to represent the behavior of customers then clustering the obtained HMMs	Behavior modeling using MSM and HMM, clustering (hierarchical and *k*-medoids)	Data of mobile Internet usage records	Mobile Internet services
(Song et al., 2016)	Proposed a methodology for segmenting customers using RFM model along with time intervals	Clustering (*k*-means), multiple corresponding analysis	RFM variables extracted from telecommunications services data	Telecommunication
(Martínez et al., 2018)	Proposed a model to predict the future purchases of customers using machine learning techniques	Logistic Lasso regression, Extreme learning machine, and gradient tree boosting	Customer purchase transaction data consists of the time and value of purchases of customers	B2B transactions of a manufacturer
(Abbasimehr & Shabani, 2019)	Modeled customer behavior as a time series of recency and monetary and using Agglomerative Clustering Algorithm	Hierarchical Agglomerative Clustering Algorithm	Recency and Monetary	Banking
(Mosaddegh et al., 2019)	Proposed an approach to understand the dynamics of customer segments	Clustering (*k*-means++ and hierarchical) frequent pattern growth (FP-Growth) algorithm	Demographic variables, Variables corresponding to customer life-time value (CLV)	Banking
(Mosaddegh et al., 2021)	Proposed a framework that segments customers based on CLV and tracks the transitions of customers between segments using association rule mining	Clustering (*k*-means and hierarchical (Agglomerative Nesting (AGNES)))	Variable utilized to quantify CLV	Banking

Spectral clustering has gained significant attention (Batista et al., 2014) due to its performance over various types of data (Filippone et al., 2008). Besides, *k*-shape is a new partitional clustering algorithm that its performance has been comprehensively validated through extensive experiments (Paparrizos & Gravano, 2015; Yang et al., 2017).

As hierarchical clustering techniques are popular and widely adopted techniques, we only explain the spectral and *k*-shape clustering methods.

3.1.1. Spectral clustering

The spectral clustering algorithm is based on graph theory and the Laplacian matrix (Ng et al., 2001; Ye & Sakurai, 2016). By considering the distance matrix $\mathbf{Dist_{n\times n}}$, which is obtained by measuring the dissimilarity between pairs of time series, the steps of spectral clustering are as follows:

1. Transform the distance matrix into a similarity matrix **S** as follows.

$$s_{ij} = \begin{cases} \exp\left(-\gamma dist_{ij}\right) & i \neq j \\ 0 & i = j, \end{cases} \tag{1}$$

where γ is the scaling factor and $dist_{ij}$ is the distance between time series i and time series j.

2. Calculate the normalized Laplacian matrix **L** using Eq. (2).

$$\mathbf{L} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} \tag{2}$$

where **D** is a $n \times n$ diagonal matrix with $d_i = \sum_{j=1}^{n} s_{ij}$

3. Compute the first k eigenvectors $(v_1, \ldots, v_k)$ of **L** and create the matrix $\mathbf{V} = (v_{ij})_{n\times k}$ that contains the first k eigenvectors as columns.

4. Obtain matrix $\mathbf{U} = (u_{ij})_{n\times k}$ by normalizing the rows of **V** using Eq. (3):

$$u_{ij} = \frac{v_{ij}}{\sqrt{\sum_j v_{ij}^2}} \tag{3}$$

5. Suppose each row of the **U** matrix as a data point $y_i \in \mathbb{R}^k$, cluster these points $(y_i)_{i=1,\ldots,n}$ using the *k*-means algorithm.

3.1.2. k-shape clustering

k-shape is a partitioning clustering algorithm that employs an iterative procedure to cluster time series data. It employs a shape-based distance measure to compute the centroid of each cluster and assign time series to each cluster. *k*-shape is a fast, domain-independent, and efficient method for time series clustering (Paparrizos & Gravano, 2015).

3.2. Distance measures

This subsection describes the distance measures utilized in this study, such as DTW, CID, $CORT$, and shape-based distance (SBD). The calculation of the distance between the two time series $X = \{x_1, x_2, \ldots, x_n\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$ using these distance measures are described in the following.

3.2.1. DTW

It computes the distance between two time series by aligning them via warping the time axis of one series on to another iteratively (Montero & Vilar, 2014). Suppose the two time series of X and Y, DTW distance can be defined by Eq. (4) as follows:

$$DTW(X,Y) = \min_{r \in M} \left(\sum_{m=1}^{M} \left| X_{im} - Y_{jm} \right| \right), \tag{4}$$

where the path element $r = (i, j)$ denotes the relationship between two series. The goal is to obtain the optimal warping path that characterizes the path that minimizes the overall distance between X and Y (Al-Naymat et al., 2009).

3.2.2. CID

Proposed by Batista et al. (2014) it exploits information about complexity differences between two time series as a correction factor for existing distance measures (for example, Euclidean distance) (Cen & Wang, 2018). A general CID measure is defined as:

$$d_{CID}(X,Y) = CF(X,Y) \cdot d(X,Y) \tag{5}$$

where $d(X,Y)$ corresponds to an existing distance measure, for example, Euclidean distance, and CF is a complexity correction factor denoted by Eq. (6):

$$CF(X,Y) = \frac{\max\{CE(X), CE(Y)\}}{\min\{CE(X), CE(Y)\}}, \tag{6}$$

where $CE(X)$ and $CE(Y)$ are the complexity estimators of X and Y, respectively. For time series X, $CE(X)$ can be calculated using Eq. (7) as:

$$CE(X) = \sqrt{\sum_{i=1}^{n-1} \left(x_i - x_{i+1} \right)^2} \tag{7}$$

3.2.3. CORT

This distance takes into account both proximity on values and on behavior when calculating the similarity between the two time series (Chouakria & Nagabhushan, 2007; Montero & Vilar, 2014). It is described by Eq. (8):

$$CORT(X,Y) = \frac{\sum_{t=1}^{n-1} \left(X_{t+1} - X_t \right) \left(Y_{t+1} - Y_t \right)}{\sqrt{\sum_{t=1}^{n-1} \left(X_{t+1} - X_t \right)^2} \sqrt{\sum_{t=1}^{n-1} \left(Y_{t+1} - Y_t \right)^2}}, \tag{8}$$

3.2.4. SBD

This is used in the k-shape clustering method described in the previous section. It is based on the normalized cross-correlation (NCC) measure (Paparrizos & Gravano, 2015, 2017). The cross-correlation measure is computed using a Fast Fourier Transformation (FFT) algorithm and normalized using the geometric mean of the autocorrelation of each individual sequence. Given two time-series vectors $\vec{X}$ and $\vec{Y}$, SBD is defined by Eq. (9) as

$$SBD(\vec{X}, \vec{Y}) = 1 - \max_{w} \left(NCC_w(\vec{X}, \vec{Y}) \right) \tag{9}$$

where w is the position that $NCC_w(\vec{X}, \vec{Y})$ is maximized.

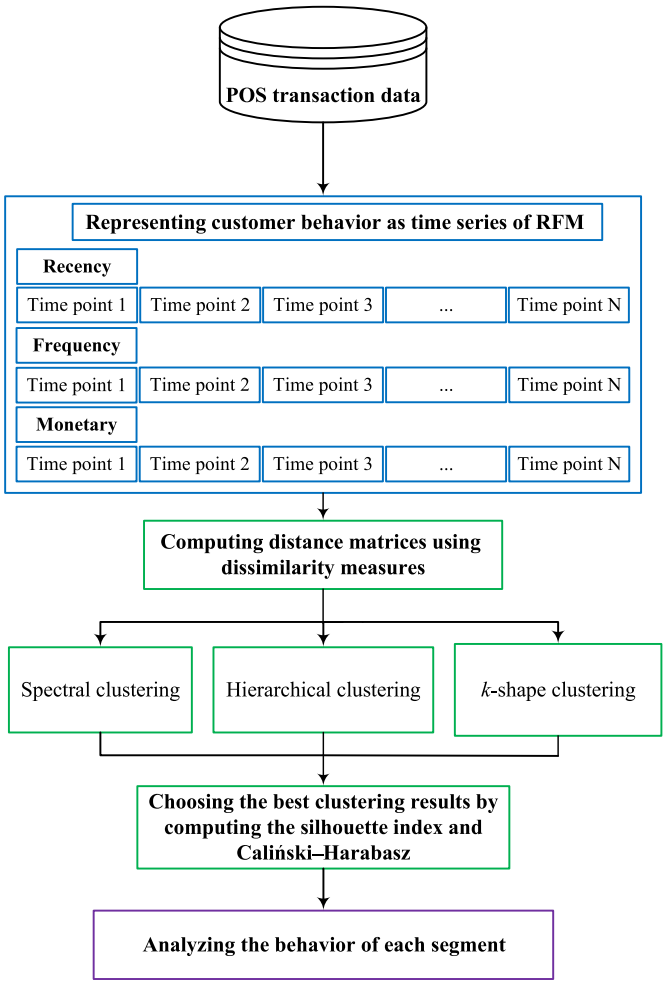


Fig. 1. The proposed framework for dynamic customer behavior.

3.3. Proposed methodology

The main objective of the proposed methodology is to provide a tool to identify different customer segments by modeling customer behavior as time series,as illustrated in Fig. 1. The proposed framework effectively integrates the RFM model with time series clustering. It extracts time series data from a stream of time-stamped transaction data to represent dynamic customer behavior. Then, it clusters the customers' time series using state-of-the-art time series clustering techniques. In the following, the detail of each step in the framework is described.

3.3.1. Representing customer behavior as time series of RFM

In this step, the original time-stamped transaction data is aggregated into bins based on a specified time interval (for example, weekly bins). Then, the time series corresponding to each variable of the RFM model is computed for each customer. The definition of time series for RFM variables are as follows:

Recency (t): The time elapsed between the end of the selected interval and the last time that the customer has a transaction in that interval.

Frequency (t): This denotes the number of transactions that were conducted using a POS device during the interval.

Monetary (t): This indicates the accumulated amount of transactions that have taken place during the interval.

Based on the purpose of a study, one can choose a particular variable to accomplish customer behavior analysis.

3.3.2. Time series clustering

The time series clustering is carried out to identify groups of customers with similar behaviors. Similar to data clustering, time series clustering consists of two critical decisions: (1) choosing a proper distance measure, (2) and selecting a clustering algorithm. Previous studies have made contrasting conclusions about the impact of each of these decisions on clustering quality. Some have indicated that distance measures play an essential role in obtaining a high-quality clustering result, while some others stated that the selection of the clustering method is also important (for example, see Batista et al. (2014) and Paparrizos and Gravano (2015)). In this step, firstly, the distance matrices of time series are computed using the corresponding distance metrics. A distance matrix contains the pairwise distance between all the time series. Four distance measures, including DTW, CID, $CORT$, and the shape-based distance (SBD), have been incorporated in the proposed methodology. Afterward, three powerful clustering methods (hierarchical, spectral, and *k*-shape) are used to preform clustering.

3.3.3. Choosing the best clustering results by computing a given cluster validity index

Once customer segmentation via different combinations of algorithms and distance metrics has been conducted, the quality of the resulted segmentation models needs to be evaluated using a suitable cluster validity index (Arbelaitz et al., 2013). Clustering is an unsupervised method with the goal of partitioning data, where the resulting clusters have high intra-cluster similarity and a low inter-cluster similarity (Arbelaitz et al., 2013). Similar to unsupervised learning tasks, in this study, the correct partition of each data point is not available so, we cannot use the external validation techniques, and the internal vitality indexes are utilized for this purpose.

3.3.4. Labeling each customer segment and analyze its behavior

In this step, each of the obtained clusters is labeled and its behavior is analyzed to discover its dominant patterns.

To summarize the main steps of the proposed methodology, we illustrate it in the form of Algorithm 1, considering the monetary variable. The algorithm takes time-stamped customer transactions as the input and creates for each customer a time series of the Monetary representing the customer's behavior over time. Then customer segmentation via time series clustering is performed. Specifically, various combinations of clustering algorithms and distance metrics are evaluated to find the best clustering model for the given cluster validity index. Finally, the obtained clusters are analyzed to reveal their dominant patterns over time.

Algorithm 1 The Proposed Methodology

Input: Time-stamped transactions of $N = \{n_1, ..., n_i\}$ customers in the format of (*CustomerID*, *TransationDate*, *Amount*), number of clusters $K = \{k_1, ..., k_m\}$, $Methods = \{Hierarchical, Spectral, k-shape\}$, $distance_metric = \{DTW, CID, CORT, SBD\}$, and time horizon T

Output: Clustering results of N customers

```
Aggregate transactions of each customer into weekly bins
Extract m(t) out of each customer n_i at each time t :∈ [1, ..., T] and create time series M_{n_i} = {m(t)}_{t=1}^{T}
for each metric_i ∈ distance_metric do
    for each n_i ∈ N do
        for each n_j ∈ N do
            D(i, j) = Distance(M_{n_i}, M_{n_j}, metric_i)
        end for
    end for
    for each k_i ∈ K do
        for each algorithm ∈ Algorithms do
            (C_1, ..., C_{k_i}) = do_clustering(algorithm, D, k_i)
            Compute the cluster validity index for (C_1, ..., C_{k_i})
        end for
    end for
    Select the best clustering model considering the computed cluster validity indexes
    Analyze the behavioral patterns of each customer segment over time and make marketing recommendations
end for
```

4. Empirical study

This section illustrates a real implementation of the proposed framework using the data acquired from a bank. We implemented the proposed framework and all the methods included in it with Python 3.7. The bank offers POS devices to merchants to process their consumers' card payments. Managing POS devices offered to merchants has cost for the bank. We analyze customers' behavior (merchants) of two guilds, including grocery and appliance retailers. The input data is the 11-month transaction data of POS devices that consists of information about POS ID, Transaction ID, and Transaction Amount. The data contains 2,156,394 transactions of purchases that have been made by consumers.

4.1. Representing customer behavior as time series of RFM

In this step, firstly, data were normalized using the min–max normalization (Han et al., 2011). Next, transaction data was aggregated into weekly bins. Then, by considering the definition of the RFM variables, the time series of R, F, and M is extracted for each week. Since all of the selected customers have transactions on all days of a week, the R attribute for all customers would be zero. So we removed R's attribute from our analysis since it has no discrimination power for behavioral segmentation. In addition, the frequency factor of each customer denotes how many transactions each customer has performed. Also, it is evident that the amount of money (monetary) of each transaction can be of any value. Therefore, the high number of transactions does not imply the high value of monetary. Consequently, for analyzing customer behavior, we chose the monetary variable as a representation of customer behavior since the ultimate goal of any firm is to gain the desired profitability (Kumar & Shah, 2004). Also, as indicated by Khobzi et al. (2014), the monetary attribute has the most importance in the banking sector.

4.2. Time series clustering

In this step, four state-of-the-art distance measures for time series clustering, namely, DTW, CID, $CORT$, and SBD, are computed for each guild. The resulting distance matrices are then fed to clustering algorithms. After computing the distance matrices, the combinations of the three incorporated clustering algorithms in the proposed framework, namely, hierarchical, spectral, and *k*-shape with the four distance measures are employed to segment customers based on the behavior similarity. For each clustering combination in this study, the number of clusters (k) is set to 4, 5, and 6.

4.3. Choosing the best clustering results by computing cluster validity indices

In this step, we compute silhouette and Caliński–Harabasz cluster validity indices to choose the optimal clustering model. According to the extensive comparative investigation of the performance of several cluster validity indices provided in Arbelaitz et al. (2013), these measures are suitable indices. We describe the silhouette and Caliński–Harabasz measures in 4.3.1 and 4.3.2, respectively, with the assumption that N data points $\{x_1, x_2, \ldots, x_N\}$ are partitioned into k clusters $\{C_1, C_2, \ldots, C_k\}$, and $x_i \in C_j$.

Table 2
The silhouette index for the clustering experiments applied to grocery retailers and appliance retailers (S: Spectral clustering, H: Hierarchical clustering)

Clustering approach	Grocery retailers			Appliance retailers		
	$k=4$	$k=5$	$k=6$	$k=4$	$k=5$	$k=6$
$S+DTW$	0.16	0.12	0.12	0.37	0.35	0.08
$S+CID$	0.38	0.34	0.32	**0.46**	0.39	0.32
$S+CORT$	0.20	0.15	0.13	0.27	0.20	0.18
$S+SBD$	0.36	0.31	0.25	0.36	0.27	0.24
$H+DTW$	0.26	0.24	0.25	0.26	0.26	0.22
$H+CID$	**0.39**	0.38	0.37	0.33	0.32	0.29
$H+CORT$	0.31	0.30	0.30	0.30	0.22	0.22
$H+SBD$	0.37	0.32	0.32	0.34	0.33	0.32
k-shape	0.22	0.22	0.22	0.25	0.26	0.25

Table 3
The Caliński–Harabasz index for the clustering experiments applied to grocery retailers and appliance retailers (S: Spectral clustering, H: Hierarchical clustering)

Clustering approach	Grocery retailers			Appliance retailers		
	$k=4$	$k=5$	$k=6$	$k=4$	$k=5$	$k=6$
$S+DTW$	80.47	68.34	58.34	121.42	95.70	78.60
$S+CID$	3581.75	3183.67	2597.61	**5696.5**	5534.25	4990.94
$S+CORT$	58.77	47.61	42.49	95.16	82.15	67.67
$S+SBD$	152.96	132.56	109.73	277.64	221.94	177.95
$H+DTW$	88.56	76.83	65.78	116.86	92.02	80.30
$H+CID$	**3641.14**	2846.47	2621.76	5125.62	4022.92	4520.98
$H+CORT$	31.98	27.42	34.75	56.97	57.61	50.54
$H+SBD$	140.02	115.38	95.86	241.39	193.32	159.15
k-shape	70.16	58.57	54.69	112.34	114.23	114.23

4.3.1. Silhouette measure

The silhouette index (SI) considers the intra-cluster and inter-cluster similarity. Let $a(x_i)$ be the average dissimilarity between x_i and all other data points in the cluster C_j, and $b(x_i)$ be the average dissimilarity between x_i and all other data points in other clusters. Also, let $|C_j|$ and $|C_l|$ be the number of points belonging to cluster C_j and C_l, respectively. Then, $a(x_i)$ and $b(x_i)$ can be obtained as follows:

$$a(x_i) = \frac{1}{|C_j| - 1} \sum_{x_j \in C_j} d(x_i, x_j), \tag{10}$$

and

$$b(x_i) = \min_{c_l \in C,\, l \neq j} \frac{1}{|C_l|} \sum_{x_j \in C_l} d(x_i, x_j) \tag{11}$$

Given the values of $a(x_i)$, $b(x_i)$, the silhouette width of x_i, $s(x_i)$, is defined as follows:

$$s(x_i) = \frac{b(x_i) - a(x_i)}{\max\{b(x_i), a(x_i)\}} \tag{12}$$

The value of $s(x_i)$ is in the range of $[-1, 1]$. By taking the average of $s(x_i)$ for all data points (Eq. (13)), the SI value can be obtained to evaluate the quality of clustering.

$$SI = \frac{1}{N} \sum_{i=1}^{N} s\left(x_i\right) \tag{13}$$

In Table 2, we summarized the SI index for the clustering approaches in grocery and appliance retailers. This index will be maximized if the clustering quality is high (Hennig et al., 2015).

4.3.2. Caliński–harabasz measure

Another cluster validity index that is used in this study is the Caliński–Harabasz (CH) index (see Caliński and Harabasz (1974)). This index has been proposed to utilize with Euclidean index, but, can be generalized to assess a clustering based on an arbitrary distance metric (Hennig & Liao, 2013). This index (Eq. (14)) is defined as the proportion of between-cluster and within-cluster variation and can be calculated as follows:

$$CH_k = \frac{N-k}{k-1} \frac{B_k}{W_k}, \quad k \neq 1 \tag{14}$$

where N is the total number of observations and k is the number of clusters formed. To evaluate a clustering based on an arbitrary dissimilarity measure, the within-cluster sum squares (W_K) (Eq. (15)) and the between-cluster sum of squares (B_K) (Eq. (16)) can be obtained as follows:

$$W_K = \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{x_i \in C_k} \sum_{x_j \in C_k} d(x_i, x_j)^2, \tag{15}$$

and

$$B_K = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{N} d(x_i, x_j)^2 - W_K. \tag{16}$$

Table 3 summarizes the CH index for the clustering approaches in grocery and appliance retailers. This index will be maximized if the clustering quality is high (Hennig et al., 2015).

According to Table 2, for grocery retailers, the highest silhouette value is achieved by employing hierarchical clustering with CID measure ($H + CID$) and setting $k = 4$. For appliance retailers, the best silhouette value is obtained by applying spectral clustering with CID measure ($S + CID$) and setting $k = 4$.

For both grocery and appliance guilds, the results demonstrated that the CID measure achieved the best clustering, which confirms that distance measure choice is more significant than the clustering algorithm (Batista et al., 2014). Also, utilizing the k-shape clustering algorithm did not yield acceptable results.

By looking at Table 3, we observe that there is a consistency between the results of clustering in terms of both SI and CH. In other words, the highest CH value is obtained by employing hierarchical clustering with CID measure ($H + CID$) and setting $k = 4$. For appliance retailers, the best CH value is yielded by using spectral clustering with CID measure ($S + CID$) and setting $k = 4$.

4.4. Labeling each customer segment and analyze its behavior

To analyze the resulting customer segments, the centroid of each segment is calculated. Each time point value of the centroid time series is computed as the arithmetic mean of all-time series' corresponding time points belonging to the segment.

4.4.1. Analysis of grocery retailers

In Figs. 2–5, the centroid series of the obtained customer segments are illustrated. As it is evident, the behavioral segmentation identified the distinguishing behavioral trends of customers effectively. To reveal the distinctive characteristics of each segment, the monetary value and the corresponding trend line for each segment are also shown. Further analysis of the characteristics of each customer segment is given in Table 4.

4.4.2. Analysis of appliance retailers

Customers of this guild are also divided into four segments (see Figs. 6–9). To represent each cluster's dominant characteristics, the monetary value (M) and the corresponding trend line for each segment are also portrayed. The characteristics and analysis of each segment are given in Table 5.

The results of the case study can be utilized by bank marketing for developing segment-specific marketing strategies to increase the value of its customers. The proposed framework completely differs from the conventional market segmentation approaches in that it accounts for the dynamic behavior of customers. The common segmentation methods often aggregate the data of customers at a one-time point, preventing useful trends from being discovered. The results of the case study demonstrate the usefulness of employing the proposed framework for analyzing customer behavior.

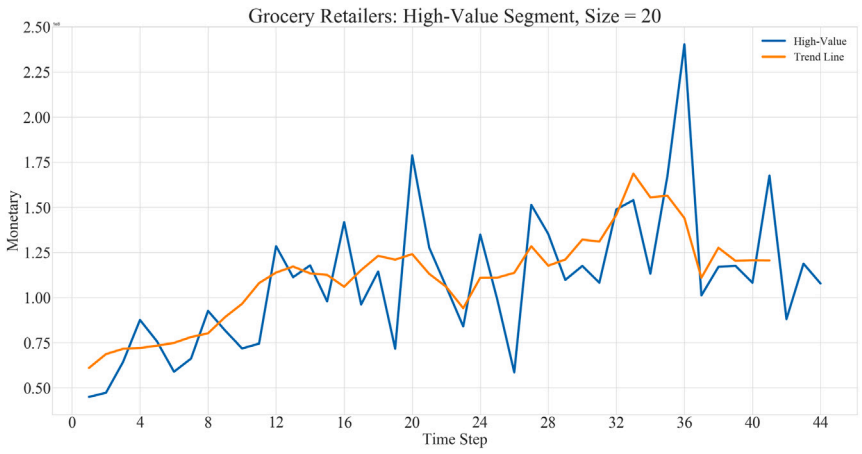


Fig. 2. The centroid series and trend of Segment 1 (High-Value) for grocery retailers.



Fig. 3. The centroid series and trend of Segment 2 (Low-Value) for grocery retailers.



Fig. 4. The centroid series and trend of Segment 3 (Middle-Value) for grocery retailers.

5. Managerial implication

5.1. Research originality

Our first contribution is the proposed framework that innovatively incorporates time dimension into the RFM model to capture customer dynamics using time series. It is entirely different from the previous studies, which aggregate the data pertinent to customer data to conduct segmentation. Also, the proposed framework enables both descriptive and predictive analysis of customer behavior. Its predictive nature is considered as the main advantage over the past studies in this context. Our second contribution is presenting a way in which spectral clustering can be adapted to time series clustering. Also, we compared the results of three state-of-the-art clustering techniques in this paper, which gives insight into the performance of these algorithms in this context. Although we applied the proposed methodology in the context of the banking industry, it can be useful for capturing and analyzing

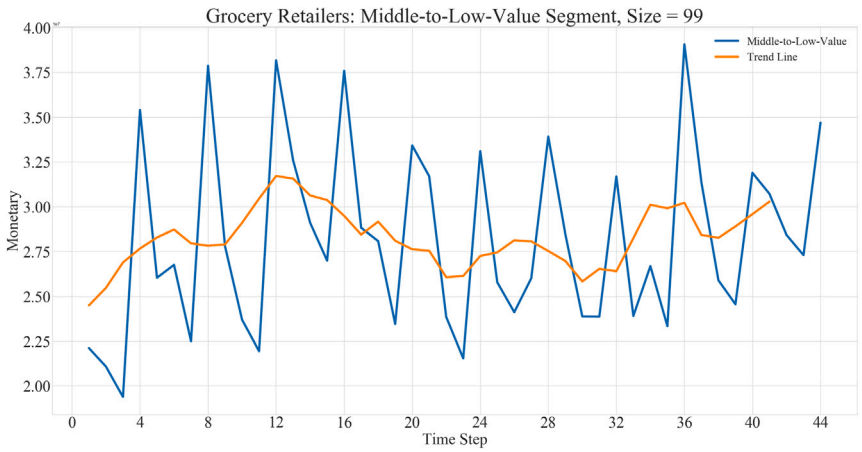


Fig. 5. The centroid series and trend of Segment 4 (Middle-to-Low-Value) for grocery retailers.

Table 4
The analysis of the grocery retailer guild.

Segment	Size	Label	Analyzes
Segment 1	20	High-Value	This segment is the most valuable in terms of monetary. This cluster (as seen in Fig. 2) exhibits seasonal behavior, and there is a high M in the first week of each month and a low M in the last week of the month. This behavior is observable during the 44 weeks and can be interpreted as the monthly seasonality. Furthermore, the overall trend is growing, which indicates that this segment will continue its profitability. The bank should devise marketing programs to maintain the profitability of this segment.
Segment 2	64	Low-Value	This segment (as seen in Fig. 3) contains the least valuable customers. It shows a strong seasonality pattern. As the trend line indicates, this cluster has a growing trend in the first 12 weeks, then its behavior becomes relatively steady, and in the last weeks, its trend starts to grow again. Because of the large number of customers, the grocery guild is considered an attractive guild to banks. One reason for being the least value segment is that other rival banks may attract these customers. Because handling each POS device has a high cost, it can decide to retain this segment or eliminate this segment from its customer base. Effective marketing program to maintain the customers of this segment and increase the value of these customers should be devised.
Segment 3	74	Middle-Value	Similar to the previous clusters, this segment (as seen in Fig. 4) shows strong seasonality. The trend is growing at the beginning of the period, then declines and becomes steady. The bank should devise marketing programs to turn these customers into high-value customers.
Segment 4	99	Middle-to-Low-Value	This segment (as seen in Fig. 5) shows patterns similar to Segments 1, 2, and 3 so that these customers showed monthly seasonality. Because of seasonality, the corresponding trend is not clear. Similar to Segment 3, the bank should design and execute specific marketing programs to prevent them from becoming low-value customers.

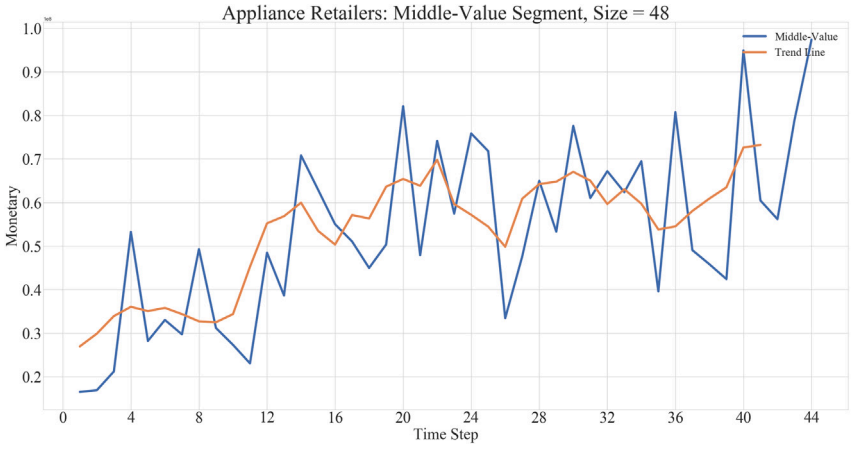


Fig. 6. The centroid series and trend of Segment 1 (Middle-Value) for appliance retailers.

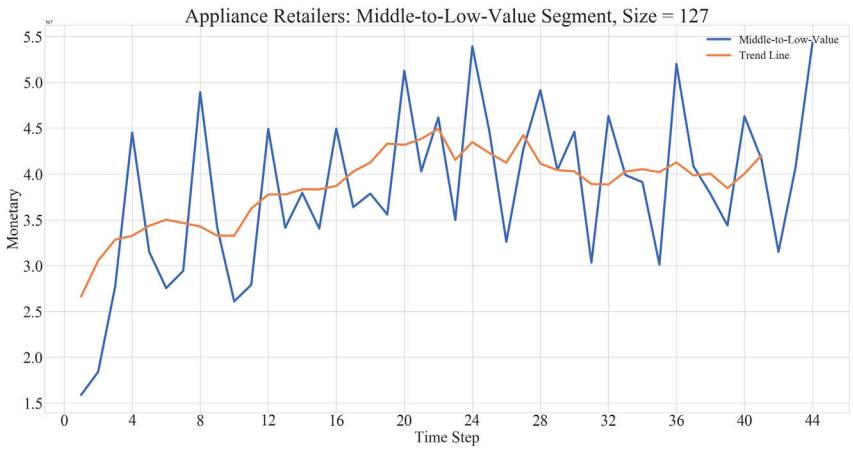


Fig. 7. The centroid series and trend of Segment 2 (Middle-to-Low-Value) for appliance retailers.



Fig. 8. The centroid series and trend of Segment 3 (High-Value) for appliance retailers.



Fig. 9. The centroid series and trend of Segment 4 (Low-Value) for appliance retailers.

customer dynamics in other sectors such as retailing, in which there are streams of time-stamped customer transactions.

5.2. Integration of the proposed framework into an information system for decision making

The proposed methodology is aimed toward transforming customers' time-stamped transactions data into time series. Assuming that the input of an information system is customer transactions data, time series analysis tools such as time series clustering, time series classification, and time series forecasting can assist managers in making decisions regarding customers throughout their life-cycle. Some applications based on the proposed approach are as follows:

1. Customer Segmentation: Analyzing different behavioral groups of customers over time by using time series clustering techniques.

Table 5
The analysis of appliances retailer guild.

Segment	Size	Label	Analyzes
Segment 1	48	Middle-Value	This segment (as seen in Fig. 6) could be interpreted as a Middle-Value cluster. The overall trend of this segment is growing. Depending on the nature of the home appliance guild, consumers may have large purchases at some time points of the year (e.g., on certain occasions). Consequently, we witness that customers (merchants) at some time points of the year have a high value of M, and at other times their M value shave declined.
Segment 2	127	Middle-to-Low-Value	This segment (as seen in Fig. 7) shows a growing trend until week 22; then its trend becomes steady. Overall, the trend is positive. Due to the large number of customers in this segment, the bank should provide an effective marketing program to maintain the customers of this segment and increase the value of these customers.
Segment 3	29	High-Value	Segment 3 (as seen in Fig. 8) is the most valuable group of customers. Strong seasonality can be observed in the plot. The trend is increasing until week 24, from then until the end of the period it is slightly declining. This behavior is different from other segments as they have an increasing trend at the end of the period and must be thoroughly investigated to identify the reasons.
Segment 4	214	Low-Value	Customers belonging to this cluster are considered as Low-Value customers. The behavior of the customers of this segment (as seen in Fig. 9 is similar to the other segments of this guild, which means that a strong seasonality is apparent in the plot. Overall, the trend is growing, indicating that these customers are active and are not prone to churn. The number of customers in this segment is slightly more than half of the total. The large size of this segment indicates that the bank should devise effective marketing programs to improve the value of these customers.

2. Churn Detection: Customer churn is a significant problem in firms. Detecting churners and adopting proper action for retention is essential (Abbasimehr et al., 2013; Alboukaey et al., 2020). Similar to customer segmentation, the majority of churn detection studies are based on static approaches. By representing the dynamic behavior of customers using time series and labeling each customer as loyal or churner, an effective churn detection tool can be implemented using a robust binary time series classifier.
3. Customer Behavior Forecasting: Customer behavior forecasting is regarded as one of the significant areas of CRM. By representing customer behavior as a time series, powerful time series forecasting methods (Abbasimehr & Paki, 2021; Abbasimehr et al., 2021; Montero-Manso & Hyndman, 2021; Yanovitzky & VanLear, 2008). The main difference of our approach compared to the previous studies is its predictive application.

5.3. Limitation

The first limitation of our work is that we only have access to customer's transaction data and not to other customer variables such as location, store size, and number of personnel for more analysis of the obtained segments.

The second limitation is that the utilized dataset consists of eleven consecutive months of customer transaction data. With the data of a longer period, we could gain a deeper insight into customer behavior.

6. Conclusion and future work

Businesses must employ tools and techniques to identify the behavioral patterns of their customers effectively. This study proposed a framework that detects dominant customer behavior patterns using time series analysis techniques. The proposed framework firstly builds a time series corresponding to each RFM variable, then, using time series clustering techniques, identifies behaviorally similar groups of customers. The proposed framework was implemented using the POS devices transaction data of a bank. Two retailer guilds, including grocery and appliance retailers, were selected. The best clustering results for grocery retailers were reached using hierarchical clustering with CID measure. Furthermore, for appliance retailers, the best segmentation in terms of the silhouette measure was obtained by applying spectral clustering with CID. The obtained segments for each guild were analyzed, and the behavioral patterns were discovered. Each guild's customer segments were labeled into four categories: High-Value, Middle-Value, Middle-to-Low-Value, and Low-Value. According to each segment's characteristics, the bank could formulate effective marketing programs to increase customers' lifetime value. For our future research, we will forecast the behavior of each segment using deep learning techniques. Also, we intend to develop a new time series averaging method to compute the centroid of clusters effectively.

CRediT authorship contribution statement

Hossein Abbasimehr: Conceptualization, Methodology, Software, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Aram Bahrini:** Data curation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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